

Overview

HIDEBAF is a feature-rich archive and encryption toolkit built around the Scorched Earth methodology. It now supports multi-token archives, optional payload compression, plugin extensions, archive verification, secure wipe workflows, auditing, and visual CLI themes.

What is HIDEBAF?

HIDEBAF stands for Highly Incubated Dragonian Encryptory Binary using an Archive of Files. It combines military-grade AES-256-GCM encryption with a flexible token model and a growing plugin ecosystem.

Key Characteristics

- **Encryption Algorithm:** AES-256-GCM with 256-bit keys
- **Archive Format:** Custom binary .HIDEBAF archives
- **Token Format:** .EBFATs token export/import files
- **Runtime:** Bun 1.3.14+ with Node-compatible modules
- **Plugin Support:** Base plugins plus custom plugin installation
- **Platform:** Windows, macOS, Linux
- **License:** MIT
- **Version:** 1.0.0

Features

Core Encryption Features

- **Military-grade encryption:** AES-256-GCM payloads with authentication tags
- **Scorched Earth methodology:** Multi-token redirection with real and decoy tokens
- **Archive hints:** Store optional metadata directly in the archive header
- **Compression control:** Leave payload compression enabled by default or disable it with `--no-compress`
- **Offline operation:** Everything runs locally without a network dependency

Token and Archive Controls

- **Multiple tokens per archive:** Create real and decoy tokens for layered access scenarios
- **Token export/import:** Build and inspect .EBFATs files from the CLI
- **Archive inspection:** View slot count, flags, hint text, and archive metadata without decrypting
- **Batch processing:** Encrypt multiple files into one archive in a single command
- **Secure file handling:** Supports large files, multiple file types, and custom output paths

Plugin & Extensibility

- **Base plugins:** Built-in modules for logging, verification, compression analysis, and banners
- **Custom plugins:** Install new plugins from disk with `plugin add`
- **Plugin lifecycle:** Inspect, remove, and query plugin metadata from the CLI
- **Hook-based design:** Plugins can respond to hide/reveal lifecycle events

Operational Utilities

- **Archive verification:** Confirm structure, flags, and checksums
- **Benchmarking:** Measure local AES-256-GCM and KDF throughput
- **Secure wipe:** Overwrite and remove files with the Scorched Earth shredder
- **Audit logs:** Inspect recent operations through the logger plugin
- **Visual themes:** Change the CLI banner experience with `theme`

Installation

Prerequisites

1. **Bun Runtime:** Version 1.3.14 or higher
2. **Operating System:** Windows 7+, macOS 10.13+, or Linux
3. **Disk Space:** Minimum 50 MB for installation
4. **RAM:** Minimum 256 MB (512 MB recommended)
5. **Terminal Access:** Command-line interface access is required

Installation Steps

1. Install Bun

```
curl -fsSL https://bun.sh/install | bash
```

2. Verify Bun

```
bun --version
```

Expect `1.3.14` or newer.

3. Open the project folder

```
cd HIDEBAF/Com.DAPOWER99.hidebaf
```

4. Install dependencies

```
bun install
```

5. Run the CLI

```
bun ./Bin/Versions/1.0/hidebaff.js --help
```

✓ **Ready to use:** HIDEBAF now supports encryption, token export/import, plugins, verification, benchmarking, and more.

Usage Guide

Basic Workflow

1. **Encrypt files:** Use `hide` to build a `.HIDEBAF` archive
2. **Create tokens:** Use `export-tokens` to build an `.EBFATs` file
3. **Decrypt archives:** Use `reveal` with a matching token
4. **Inspect or verify:** Use `info`, `verify`, or `compress-stats`

Example: Encrypt with compression and a hint

```
$ bun ./Bin/Versions/1.0/hidebaff.js hide report.pdf --hint "finance"
```

```
✓ Archive created successfully  
✓ Tokens exported: mytokens.EBFATs
```

Example: Add decoy tokens

```
$ bun ./Bin/Versions/1.0/hidebaff.js hide secret.txt --decoy decoy-on
```

Example: Reveal an archive

```
$ bun ./Bin/Versions/1.0/hidebaff.js reveal report.HIDEBAF --token "m
```

Example: Inspect plugins

```
$ bun ./Bin/Versions/1.0/hidebaff.js plugin list
```

Example: Verify integrity

```
$ bun ./Bin/Versions/1.0/hidebaff.js verify report.HIDEBAF
```

Command Reference

hide [options] <files...>

Description: Encrypt one or more files into a .HIDEBAF archive.

```
hidebaf hide [options] file1.txt file2.pdf
```

Options:

- `-o, --output <file>` : Output archive path
- `--token <token>` : Provide a real token non-interactively
- `--hint <text>` : Store a plaintext hint in the archive header
- `--no-compress` : Disable zlib payload compression
- `--decoy <password>` : Add a decoy token; repeatable
- `--export-tokens <path>` : Export all tokens to a .EBFATs file

reveal [options] <archive>

Description: Decrypt a .HIDEBAF archive using a token.

```
hidebaf reveal [options] archive.HIDEBAF
```

export-tokens [options]

Description: Interactively build and export a .EBFATs file.

```
hidebaf export-tokens --output tokens.EBFATs
```

import-tokens <ebfats>

Description: List the token labels and types stored in a .EBFATs file.

```
hidebaf import-tokens tokens.EBFATs
```

info <archive>

Description: Display archive metadata without decrypting the contents.

```
hidebaf info archive.HIDEBAF
```

plugin <subcommand>

Description: List, install, remove, or inspect plugins.

```
hidebaf plugin list
hidebaf plugin add ./my-plugin.js
hidebaf plugin remove my-plugin
hidebaf plugin info verifier
```

verify <archive>

Description: Cryptographically verify archive structure and checksums.

```
hidebaf verify archive.HIDEBAF
```

benchmark

Description: Measure local AES-256-GCM and scrypt KDF throughput.

```
hidebaf benchmark
```

wipe <files...>

Description: Securely overwrite and delete files using the Scorched Earth shredder.

```
hidebaf wipe secret.txt
```

audit

Description: Show recent operation logs from the logger plugin.

```
hidebaf audit
```

compress-stats <archive>

Description: Analyze compression ratio and space savings for an archive.

```
hidebaf compress-stats archive.HIDEBAF
```

theme [themeName]

Description: View or set the CLI banner theme.

```
hidebaf theme
```

```
hidebaf theme cyber
```

Advanced Capabilities

Plugin-ready workflow: The CLI now supports a plugin lifecycle, base plugins, and custom plugin installation under the project Lib folders.

- **Compression analysis:** Measure payload savings with compress-stats.
- **Archive integrity verification:** Validate archive structure and checksums with verify.
- **Performance benchmarking:** Check local AES-256-GCM and KDF throughput with benchmark.
- **Secure deletion:** Use wipe for irreversible file removal.
- **Audit visibility:** Review recent logs with audit.
- **Visual presentation:** Toggle between dragon, stealth, cyber, and classic themes.

System Requirements

Minimum Requirements

- Processor: 1.0 GHz 64-bit
- RAM: 256 MB
- Disk Space: 50 MB
- Operating System: Windows 7+ / macOS 10.13+ / Linux

Runtime Requirements

- Bun Runtime: 1.3.14 or higher
- Purpose: JavaScript/TypeScript runtime and package manager

Frequently Asked Questions

What is the encryption strength?

HIDEBAF uses AES-256-GCM and is designed for strong local encryption with authenticated payloads. The implementation is suitable for personal and professional file protection.

Can I decrypt my archive without the token?

No. Access to the archive still depends on the correct token. The archive metadata can be inspected without decryption, but the contents remain protected.

Can I use decoy tokens?

Yes. HIDEBAF supports real and decoy tokens so you can create layered access scenarios without exposing a single master token.

Can I add plugins?

Yes. The CLI includes plugin management commands so you can inspect built-in plugins or install custom ones from disk.

Can I verify an archive after creation?

Yes. Use `verify` to inspect archive integrity and checksum-related metadata.

Can I disable compression?

Yes. Use `--no-compress` when creating archives if you want to avoid zlib payload compression.

Support & Resources

- **Documentation:** Review the markdown files in the package and the built-in CLI help
- **Built-in Help:** `hidebaf help [command]`
- **GitHub Issues:** <https://github.com/DAPOWER99/HIDEBAF/issues>
- **GitHub Discussions:** <https://github.com/DAPOWER99/HIDEBAF/discussions>

HIDEBAF v1.0.0

Status: Stable CLI release

License: MIT

Bun Requirement: 1.3.14+